

CODECHECK certificate 2026-025

<https://doi.org/10.5281/zenodo.22935939>



Table 1: CODECHECK summary

Item	Value
Title	<i>Constructing melanopic light exposure profiles from sparse daily-route measurements</i>
Author(s)	Niloufar Tabandeh  , Jennifer A. Veitch  , Manuel Spitschan 
Reference	https://doi.org/10.1038/s44323-026-00108-x
Repository	https://github.com/codecheckers/TabandehEtAl_npjBiolTimingSleep_2026
Codechecker(s)	Bruno de Paula Kinoshita 
Date of check	2026-09-20
Summary	Reproduction using the Workflow hosted at https://github.com/tscnlab/TabandehEtAl_npjBiolTimingSleep_2026 , using the Git commit <code>e4a1b5500394de6e317aabd8315df60b8bedc6e3</code> . Installed the dependencies on a Linux Ubuntu 26.04, using micromamba, with R 4.6.1, latest quarto and renv, following the instructions from the project README file. There were minor issues during the dependencies installation that authors started working on improvements for the documentation. A Jupyter notebook was used, as instructed in the project README, to execute part of the workflow. The actual data processing was verified using the R Quarto notebook, that took about 11 hours to complete. After that, the remaining steps of the Jupyter notebook were executed, and produced the same figures and table data used in the article. The data, plots, tables, all looked correct.

Table 2: Summary of output files generated

File	Comment	Size (b)
<code>terminal_output.log</code>	The terminal output of executing the R script on my personal laptop.	786
<code>LightLogR.html</code>	Updated notebook/document after re-running it locally.	1443873
<code>trajectory.ipynb</code>	Updated notebook/document after re-running it locally.	2679404

File	Comment	Size (b)
datatree.txt	Generated directory tree, produced after running the scripts locally.	9096
Figure1.pdf	Plot recreated locally with Python/R.	1093111
Figure1.png	Plot recreated locally with Python/R.	1119298
Figure10.pdf	Plot recreated locally with Python/R.	25387
Figure10.png	Plot recreated locally with Python/R.	439586
Figure11.pdf	Plot recreated locally with Python/R.	41316
Figure11.png	Plot recreated locally with Python/R.	130453
Figure12.pdf	Plot recreated locally with Python/R.	16802
Figure12.png	Plot recreated locally with Python/R.	402564
Figure2.pdf	Plot recreated locally with Python/R.	33834
Figure2.png	Plot recreated locally with Python/R.	173670
Figure9.pdf	Plot recreated locally with Python/R.	371600
Figure9.png	Plot recreated locally with Python/R.	374107
Models_distribution_grid.png	Plot recreated locally with Python/R.	666654
activity_block_matched.pdf	Plot recreated locally with Python/R.	44083
activity_block_matched.png	Plot recreated locally with Python/R.	343978
broad_context_matched.pdf	Plot recreated locally with Python/R.	43471
broad_context_matched.png	Plot recreated locally with Python/R.	349142
exact_slot_matched.pdf	Plot recreated locally with Python/R.	42858
exact_slot_matched.png	Plot recreated locally with Python/R.	350369
indoor_only_bound.pdf	Plot recreated locally with Python/R.	41411
indoor_only_bound.png	Plot recreated locally with Python/R.	284753
melanopic_trajectory_days.png	Plot recreated locally with Python/R.	324730
outdoor_only_bound.pdf	Plot recreated locally with Python/R.	39583
outdoor_only_bound.png	Plot recreated locally with Python/R.	269183
photopic_trajectory_days.png	Plot recreated locally with Python/R.	353842
tailored_model_dose_plot.png	Plot recreated locally with Python/R.	198982
tailored_model_effect_plot.png	Plot recreated locally with Python/R.	375954
unrestricted_pooled.pdf	Plot recreated locally with Python/R.	43146
unrestricted_pooled.png	Plot recreated locally with Python/R.	294398
activity_block_matched_melanopic.csv	CSV recreated locally with Python/R.	7132942
activity_block_matched_melanopic_numerical.csv	CSV recreated locally with Python/R.	3232088

File	Comment	Size (b)
<code>broad_context_matched_melanopic.csv</code>	CSV recreated locally with Python/R.	7135650
<code>broad_context_matched_melanopic_numeric.csv</code>	CSV recreated locally with Python/R.	3235735
<code>exact_slot_matched_melanopic.csv</code>	CSV recreated locally with Python/R.	7110541
<code>exact_slot_matched_melanopic_numeric.csv</code>	CSV recreated locally with Python/R.	3208657
<code>indoor_only_bound_melanopic.csv</code>	CSV recreated locally with Python/R.	5542524
<code>indoor_only_bound_melanopic_numeric.csv</code>	CSV recreated locally with Python/R.	2842251
<code>outdoor_only_bound_melanopic.csv</code>	CSV recreated locally with Python/R.	7284558
<code>outdoor_only_bound_melanopic_numeric.csv</code>	CSV recreated locally with Python/R.	3384596
<code>unrestricted_pooled_melanopic.csv</code>	CSV recreated locally with Python/R.	7112323
<code>unrestricted_pooled_melanopic_numeric.csv</code>	CSV recreated locally with Python/R.	3222232
<code>pivot_melanopic_metadata_trajectory_canada.csv</code>	CSV recreated locally with Python/R.	3069
<code>pivot_photopic_melanopic_trajectory_canada.csv</code>	CSV recreated locally with Python/R.	1234
<code>pivot_summary_metrics_stats.csv</code>	CSV recreated locally with Python/R.	1523
<code>pivot_summary_trajectory_canada.csv</code>	CSV recreated locally with Python/R.	96
<code>activity_block_matched_melanopic_sequence.csv</code>	CSV recreated locally with Python/R.	15013669
<code>broad_context_matched_melanopic_sequence.csv</code>	CSV recreated locally with Python/R.	15058456
<code>exact_slot_matched_melanopic_sequence.csv</code>	CSV recreated locally with Python/R.	14848503
<code>indoor_only_bound_melanopic_sequence.csv</code>	CSV recreated locally with Python/R.	13798244
<code>outdoor_only_bound_melanopic_sequence.csv</code>	CSV recreated locally with Python/R.	17734514
<code>unrestricted_pooled_melanopic_sequence.csv</code>	CSV recreated locally with Python/R.	14925163

Summary

Reproduction using the Workflow hosted at https://github.com/tscnlab/TabandehEtAl_npjBiolTimin gSleep_2026, using the Git commit e4a1b5500394de6e317aabd8315df60b8bedc6e3. Installed the dependencies on a Linux Ubuntu 26.04, using micromamba, with R 4.6.1, latest quarto and renv, following the instructions from the project README file. There were minor issues during the dependencies installation that authors started working on improvements for the documentation. A Jupyter notebook was used, as instructed in the project README, to execute part of the workflow. The actual data processing was verified using the R Quarto notebook, that took about 11 hours to complete. After that, the remaining steps of the Jupyter notebook were executed, and produced the same figures and table data used in the article. The data, plots, tables, all looked correct.

CODECHECKER notes

A few dependencies had to be manually installed on Ubuntu Linux so that renv succeeded in installing the R environment. Authors helped to identify the issues, and commented they would address the installation instructions. The processing time was also not documented, but it matched the time informed by authors afterwards, of approximately eleven hours. The output of the R Quarto document produce violin plots that seemed a bit different from those from the paper. However, executing the rest of the Jupyter notebook, the same plots are produced in Python and these look identical to the ones from the paper. No other issues found.

Recommendations to the authors

- Update the installation instructions;
- Mention that some files are pre-generated with a different version from the final used in the paper;
- Include a continuous integration test pipeline, so readers/users can use that as reference.

Manifest files

CSV files

results/tables/LightLogR/metrics_raw/activity_block_matched_melanopic.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	7	23400s (~6.5 hours)	16
3	50	10	19800s (~5.5 hours)	13
4	50	7	7200s (~2 hours)	18

results/tables/LightLogR/metrics_raw/activity_block_matched_melanopic_numeric.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	7	23400	16
3	50	10	19800	13
4	50	7	7200	18

results/tables/LightLogR/metrics_raw/broad_context_matched_melanopic.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	10	25200s (~7 hours)	13
3	50	8	19800s (~5.5 hours)	14
4	50	9	7200s (~2 hours)	11

results/tables/LightLogR/metrics_raw/broad_context_matched_melanopic_numeric.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1

	count	unique	top	freq
2	50	10	25200	13
3	50	8	19800	14
4	50	9	7200	11

results/tables/LightLogR/metrics_raw/exact_slot_matched_melanopic.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	7	25200s (~7 hours)	15
3	50	9	19800s (~5.5 hours)	16
4	50	7	5400s (~1.5 hours)	15

results/tables/LightLogR/metrics_raw/exact_slot_matched_melanopic_numeric.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	7	25200	15
3	50	9	19800	16
4	50	7	5400	15

results/tables/LightLogR/metrics_raw/indoor_only_bound_melanopic.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	10	18000s (~5 hours)	11
3	50	9	9000s (~2.5 hours)	11
4	50	2	0s	49

results/tables/LightLogR/metrics_raw/indoor_only_bound_melanopic_numeric.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	10	18000	11
3	50	9	9000	11
4	50	2	0	49

results/tables/LightLogR/metrics_raw/outdoor_only_bound_melanopic.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	2	39600s (~11 hours)	49
3	50	2	39600s (~11 hours)	49
4	50	10	25200s (~7 hours)	11

results/tables/LightLogR/metrics_raw/outdoor_only_bound_melanopic_numeric.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	2	39600	49
3	50	2	39600	49
4	50	10	25200	11

results/tables/LightLogR/metrics_raw/unrestricted_pooled_melanopic.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	11	21600s (~6 hours)	11
3	50	10	19800s (~5.5 hours)	10
4	50	9	9000s (~2.5 hours)	14

results/tables/LightLogR/metrics_raw/unrestricted_pooled_melanopic_numeric.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	50	50	Sequence	1
1	50	50	dose	1
2	50	11	21600	11
3	50	10	19800	10
4	50	9	9000	14

results/tables/pivot_melanopic_metadata_trajectory_canada.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	24	24	time	1
1	25	25	2023-06-14	1
2	25	5	Indoor (mixed)	11
3	25	10	Indoor (office)	9
4	23	23	2023-06-16	1
5	23	6	Indoor (electriclight)	9
6	23	8	Indoor (office)	9
7	21	21	2023-06-19	1
8	21	5	Indoor (daylight)	14
9	21	9	Indoor (office)	9

results/tables/pivot_photopic_melanopic_trajectory_canada.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	24	24	time	1
1	25	25	Photopic illuminance (lx)	1
2	23	23	Photopic illuminance (lx)	1
3	21	21	Photopic illuminance (lx)	1
4	25	25	Melanopic EDI (lx)	1
5	23	23	Melanopic EDI (lx)	1
6	21	21	Melanopic EDI (lx)	1

results/tables/pivot_summary_metrics_stats.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	8	8	metric	1
1	8	8	dose	1
2	8	8	dose	1
3	8	8	dose	1
4	8	8	dose	1
5	8	8	dose	1
6	8	6	14400.0	3
7	8	6	18000.0	3
8	8	5	23400.0	4
9	8	7	30600.0	2
10	8	6	34200.0	2
11	8	7	10800.0	2
12	8	6	14400.0	3
13	8	6	19800.0	2
14	8	7	25200.0	2
15	8	7	32400.0	2
16	8	5	0.0	4
17	8	6	3600.0	3
18	8	5	7200.0	4
19	8	7	10800.0	2
20	8	7	14400.0	2

results/tables/pivot_summary_trajectory_canada.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	5	5	date	1
1	5	5	indoor	1
2	5	4	7	2
3	5	5	sum	1

results/tables/sequences/activity_block_matched_melanopic_sequence.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	22	22.0000	MH01	1.0000
1	23	20.0000	320.2	2.0000
2	23	23.0000	seq_00002	1.0000
3	23	19.0000	11466.1	2.0000
4	23	19.0000	11539.1	2.0000

	count	unique	top	freq
5	23	22.0000	845.3	2.0000
6	23	21.0000	10942.8	2.0000
7	23	21.0000	23.0	2.0000
8	23	21.0000	19.0	2.0000
9	23	21.0000	11539.1	2.0000
10	23	22.0000	24.2	2.0000
11	23	22.0000	1408.0	2.0000
12	23	21.0000	269.7	2.0000
13	23	23.0000	seq_00013	1.0000
14	23	22.0000	7783.4	2.0000
15	23	20.0000	13535.0	2.0000
16	23	22.0000	24.4	2.0000
17	23	23.0000	seq_00017	1.0000
18	23	19.0000	4227.6	2.0000
19	23	20.0000	81.1	2.0000
20	23	21.0000	6252.4	3.0000
21	23	23.0000	seq_00021	1.0000
22	23	21.0000	13535.0	2.0000
23	23	22.0000	845.3	2.0000
24	23	23.0000	seq_00024	1.0000
25	23	20.0000	3527.1	2.0000
26	23	21.0000	93.5	2.0000
27	23	20.0000	11466.1	2.0000
28	23	20.0000	14287.4	3.0000
29	23	19.0000	92.0	2.0000
30	23	19.0000	93.5	2.0000
31	23	22.0000	93.2	2.0000
32	23	22.0000	7709.9	2.0000
33	23	21.0000	11466.1	2.0000
34	23	20.0000	13503.6	2.0000
35	23	18.0000	378.2	3.0000
36	23	18.0000	185.7	3.0000
37	23	21.0000	100.5	2.0000
38	23	19.0000	19.0	2.0000
39	23	23.0000	seq_00039	1.0000
40	23	20.0000	93.2	3.0000
41	23	23.0000	seq_00041	1.0000
42	23	21.0000	269.7	2.0000
43	23	18.0000	94.3	2.0000
44	23	21.0000	248.2	2.0000
45	23	21.0000	14287.4	2.0000
46	23	22.0000	10942.8	2.0000

	count	unique	top	freq
47	23	21.0000	1185.9	2.0000
48	23	22.0000	2715.7	2.0000
49	23	21.0000	3421.9	2.0000

results/tables/sequences/broad_context_matched_melanopic_sequence.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	22	22.0000	MH01	1.0000
1	23	18.0000	11466.1	3.0000
2	23	21.0000	99.9	2.0000
3	23	22.0000	245.8	2.0000
4	23	21.0000	378.2	2.0000
5	23	23.0000	seq_00005	1.0000
6	23	20.0000	13503.6	2.0000
7	23	20.0000	6252.4	2.0000
8	23	22.0000	24.4	2.0000
9	23	21.0000	378.2	2.0000
10	23	20.0000	14938.9	2.0000
11	23	22.0000	24.4	2.0000
12	23	21.0000	207.8	2.0000
13	23	21.0000	7709.9	2.0000
14	23	22.0000	2715.7	2.0000
15	23	22.0000	100.6	2.0000
16	23	20.0000	94.3	2.0000
17	23	23.0000	seq_00017	1.0000
18	23	21.0000	11539.1	2.0000
19	23	23.0000	seq_00019	1.0000
20	23	23.0000	seq_00020	1.0000
21	23	22.0000	11539.1	2.0000
22	23	23.0000	seq_00022	1.0000
23	23	21.0000	100.5	2.0000
24	23	19.0000	207.8	2.0000
25	23	21.0000	3858.5	2.0000
26	23	22.0000	3421.9	2.0000
27	23	20.0000	3421.9	2.0000
28	23	21.0000	19.0	2.0000
29	23	22.0000	286.4	2.0000
30	23	22.0000	286.4	2.0000
31	23	22.0000	93.5	2.0000
32	23	20.0000	7678.2	2.0000

	count	unique	top	freq
33	23	21.0000	93.9	3.0000
34	23	20.0000	8660.9	2.0000
35	23	20.0000	93.5	2.0000
36	23	20.0000	7709.9	2.0000
37	23	23.0000	seq_00037	1.0000
38	23	21.0000	13503.6	2.0000
39	23	22.0000	4171.3	2.0000
40	23	21.0000	14287.4	2.0000
41	23	23.0000	seq_00041	1.0000
42	23	21.0000	320.2	2.0000
43	23	21.0000	378.2	2.0000
44	23	22.0000	47.1	2.0000
45	23	22.0000	4171.3	2.0000
46	23	23.0000	seq_00046	1.0000
47	23	22.0000	11466.1	2.0000
48	23	20.0000	94.5	2.0000
49	23	21.0000	97.9	2.0000

results/tables/sequences/exact_slot_matched_melanopic_sequence.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	22	22	MH01	1
1	23	23	seq_0001	1
2	23	23	seq_0002	1
3	23	23	seq_0003	1
4	23	23	seq_0004	1
5	23	23	seq_0005	1
6	23	23	seq_0006	1
7	23	23	seq_0007	1
8	23	23	seq_0008	1
9	23	23	seq_0009	1
10	23	23	seq_0010	1
11	23	23	seq_0011	1
12	23	23	seq_0012	1
13	23	23	seq_0013	1
14	23	23	seq_0014	1
15	23	23	seq_0015	1
16	23	23	seq_0016	1
17	23	23	seq_0017	1
18	23	23	seq_0018	1

	count	unique	top	freq
19	23	23	seq_0019	1
20	23	23	seq_0020	1
21	23	23	seq_0021	1
22	23	23	seq_0022	1
23	23	23	seq_0023	1
24	23	23	seq_0024	1
25	23	23	seq_0025	1
26	23	23	seq_0026	1
27	23	23	seq_0027	1
28	23	23	seq_0028	1
29	23	23	seq_0029	1
30	23	23	seq_0030	1
31	23	23	seq_0031	1
32	23	23	seq_0032	1
33	23	23	seq_0033	1
34	23	23	seq_0034	1
35	23	23	seq_0035	1
36	23	23	seq_0036	1
37	23	23	seq_0037	1
38	23	23	seq_0038	1
39	23	23	seq_0039	1
40	23	23	seq_0040	1
41	23	23	seq_0041	1
42	23	23	seq_0042	1
43	23	23	seq_0043	1
44	23	23	seq_0044	1
45	23	23	seq_0045	1
46	23	23	seq_0046	1
47	23	23	seq_0047	1
48	23	23	seq_0048	1
49	23	23	seq_0049	1

results/tables/sequences/indoor_only_bound_melanopic_sequence.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	22	22.0000	MH01	1.0000
1	23	19.0000	100.5	2.0000
2	23	18.0000	245.8	3.0000
3	23	20.0000	24.4	2.0000
4	23	21.0000	93.2	2.0000

	count	unique	top	freq
5	23	20.0000	248.2	3.0000
6	23	21.0000	46.3	2.0000
7	23	22.0000	93.9	2.0000
8	23	19.0000	4171.3	2.0000
9	23	20.0000	286.4	3.0000
10	23	20.0000	20.4	2.0000
11	23	20.0000	245.8	3.0000
12	23	19.0000	19.0	2.0000
13	23	21.0000	3527.1	2.0000
14	23	22.0000	551.7	2.0000
15	23	20.0000	4130.5	2.0000
16	23	20.0000	6252.4	3.0000
17	23	22.0000	19.0	2.0000
18	23	20.0000	17.5	3.0000
19	23	22.0000	93.5	2.0000
20	23	21.0000	378.2	2.0000
21	23	21.0000	286.4	2.0000
22	23	22.0000	7325.8	2.0000
23	23	20.0000	269.7	2.0000
24	23	20.0000	269.7	2.0000
25	23	20.0000	286.4	2.0000
26	23	21.0000	99.9	2.0000
27	23	20.0000	185.7	2.0000
28	23	19.0000	3858.5	2.0000
29	23	21.0000	245.8	2.0000
30	23	20.0000	70.8	3.0000
31	23	20.0000	100.6	2.0000
32	23	19.0000	81.1	2.0000
33	23	21.0000	24.4	2.0000
34	23	18.0000	1185.9	3.0000
35	23	17.0000	24.2	2.0000
36	23	20.0000	22.2	2.0000
37	23	21.0000	70.8	2.0000
38	23	21.0000	551.7	2.0000
39	23	19.0000	551.7	2.0000
40	23	20.0000	100.5	2.0000
41	23	20.0000	1185.9	3.0000
42	23	17.0000	94.3	2.0000
43	23	22.0000	23.0	2.0000
44	23	20.0000	6155.2	3.0000
45	23	20.0000	185.7	3.0000
46	23	20.0000	94.5	2.0000

	count	unique	top	freq
47	23	23.0000	seq_0047	1.0000
48	23	22.0000	94.5	2.0000
49	23	22.0000	19.0	2.0000

results/tables/sequences/outdoor_only_bound_melanopic_sequence.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

	count	unique	top	freq
0	22	22.0000	MH01	1.0000
1	23	16.0000	1408.0	3.0000
2	23	20.0000	13535.0	2.0000
3	23	16.0000	7783.4	3.0000
4	23	16.0000	7709.9	3.0000
5	23	19.0000	13535.0	3.0000
6	23	19.0000	11466.1	3.0000
7	23	16.0000	7783.4	4.0000
8	23	20.0000	10942.8	2.0000
9	23	17.0000	14772.3	3.0000
10	23	18.0000	10945.2	3.0000
11	23	17.0000	13503.6	4.0000
12	23	16.0000	14287.4	4.0000
13	23	17.0000	4227.6	3.0000
14	23	17.0000	14421.8	3.0000
15	23	20.0000	14938.9	3.0000
16	23	18.0000	4439.6	2.0000
17	23	17.0000	7783.4	3.0000
18	23	18.0000	10942.8	3.0000
19	23	19.0000	10945.2	4.0000
20	23	18.0000	4227.6	2.0000
21	23	17.0000	7709.9	2.0000
22	23	18.0000	10942.8	3.0000
23	23	16.0000	1408.0	3.0000
24	23	15.0000	5553.3	2.0000
25	23	18.0000	7709.9	4.0000
26	23	17.0000	2715.7	2.0000
27	23	19.0000	2715.7	2.0000
28	23	15.0000	14938.9	3.0000
29	23	18.0000	14287.4	3.0000
30	23	16.0000	14421.8	4.0000
31	23	20.0000	11539.1	2.0000
32	23	17.0000	5553.3	4.0000

	count	unique	top	freq
33	23	19.0000	14421.8	2.0000
34	23	17.0000	5553.3	3.0000
35	23	17.0000	10945.2	3.0000
36	23	18.0000	13503.6	2.0000
37	23	17.0000	7709.9	2.0000
38	23	19.0000	13503.6	2.0000
39	23	16.0000	7709.9	3.0000
40	23	16.0000	11539.1	3.0000
41	23	14.0000	11539.1	3.0000
42	23	17.0000	11466.1	3.0000
43	23	18.0000	33750.6	4.0000
44	23	21.0000	7783.4	3.0000
45	23	16.0000	14421.8	3.0000
46	23	15.0000	1408.0	3.0000
47	23	16.0000	11539.1	4.0000
48	23	18.0000	33750.6	3.0000
49	23	18.0000	14421.8	3.0000

results/tables/sequences/unrestricted_pooled_melanopic_sequence.csv

Author comment: *CSV recreated locally with Python/R.*

Column summary statistics:

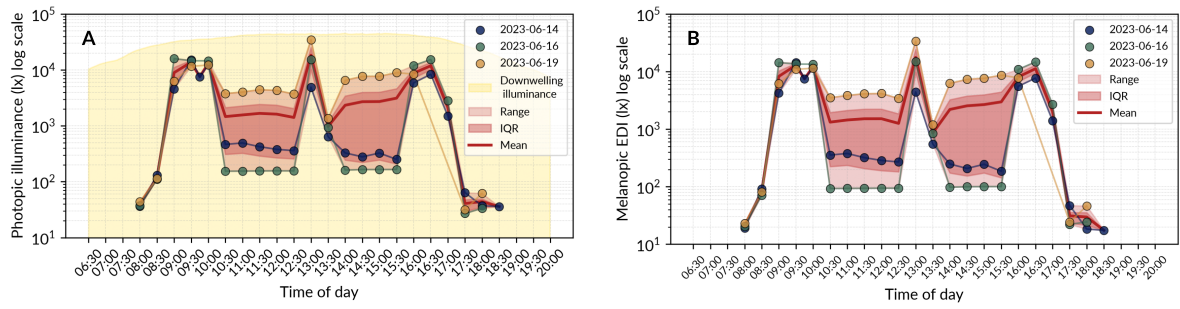
	count	unique	top	freq
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1	23	21.0000	13535.0	2.0000
2	23	20.0000	100.6	2.0000
3	23	22.0000	7709.9	2.0000
4	23	21.0000	93.5	2.0000
5	23	22.0000	22.2	2.0000
6	23	20.0000	11466.1	2.0000
7	23	20.0000	6252.4	2.0000
8	23	22.0000	33750.6	2.0000
9	23	23.0000	seq_0009	1.0000
10	23	20.0000	47.1	2.0000
11	23	21.0000	94.5	2.0000
12	23	21.0000	355.0	3.0000
13	23	21.0000	185.7	2.0000
14	23	20.0000	7474.6	3.0000
15	23	22.0000	19.0	2.0000
16	23	21.0000	11466.1	2.0000
17	23	23.0000	seq_0017	1.0000
18	23	22.0000	46.3	2.0000

	count	unique	top	freq
19	23	22.0000	320.2	2.0000
20	23	20.0000	81.1	2.0000
21	23	21.0000	94.5	2.0000
22	23	20.0000	4130.5	2.0000
23	23	22.0000	24.4	2.0000
24	23	21.0000	24.2	2.0000
25	23	23.0000	seq__0025	1.0000
26	23	22.0000	3858.5	2.0000
27	23	22.0000	551.7	2.0000
28	23	20.0000	355.0	3.0000
29	23	22.0000	378.2	2.0000
30	23	22.0000	14287.4	2.0000
31	23	22.0000	378.2	2.0000
32	23	20.0000	14938.9	2.0000
33	23	20.0000	24.4	2.0000
34	23	21.0000	286.4	2.0000
35	23	21.0000	14421.8	2.0000
36	23	22.0000	551.7	2.0000
37	23	22.0000	14287.4	2.0000
38	23	20.0000	8660.9	2.0000
39	23	21.0000	1185.9	2.0000
40	23	21.0000	94.5	2.0000
41	23	20.0000	7783.4	2.0000
42	23	21.0000	13503.6	2.0000
43	23	21.0000	20.4	2.0000
44	23	22.0000	94.5	2.0000
45	23	20.0000	100.6	2.0000
46	23	21.0000	100.6	3.0000
47	23	21.0000	185.7	2.0000
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49	23	21.0000	93.2	2.0000

Figures

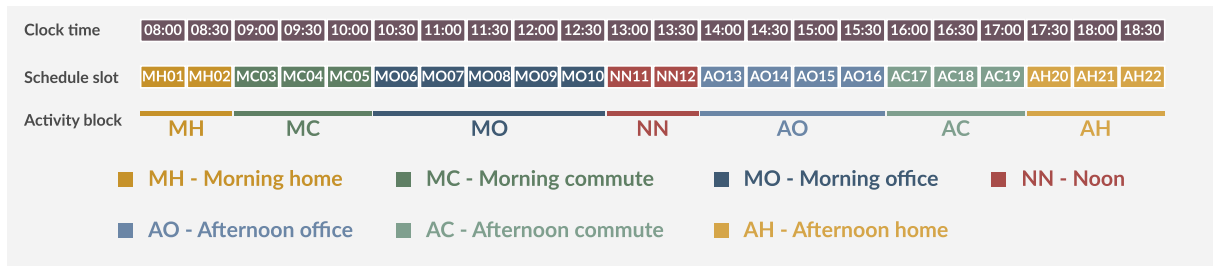
results/plots/Figure1.pdf

Author comment: *Plot recreated locally with Python/R.*



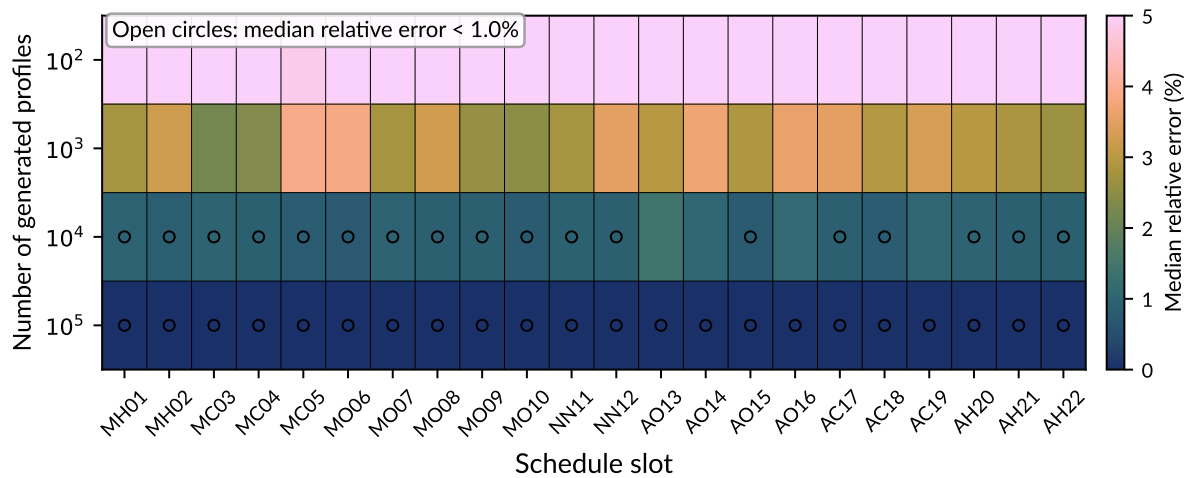
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Author comment: *Plot recreated locally with Python/R.*



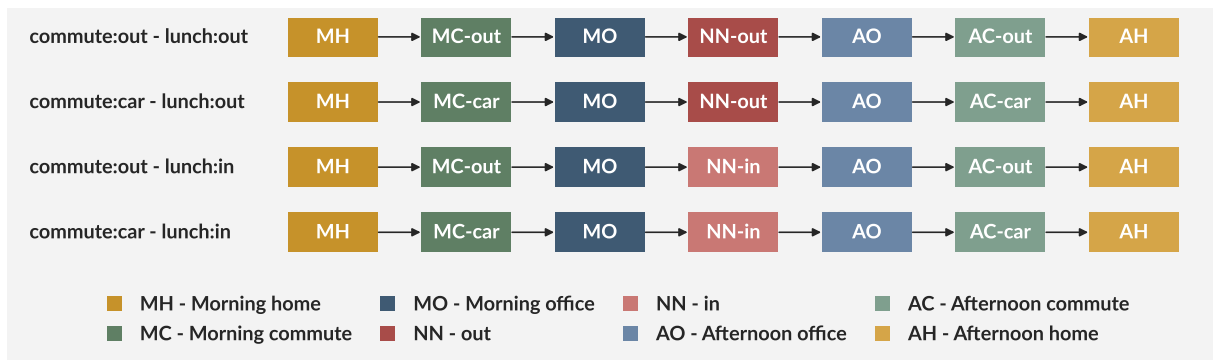
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Author comment: *Plot recreated locally with Python/R.*



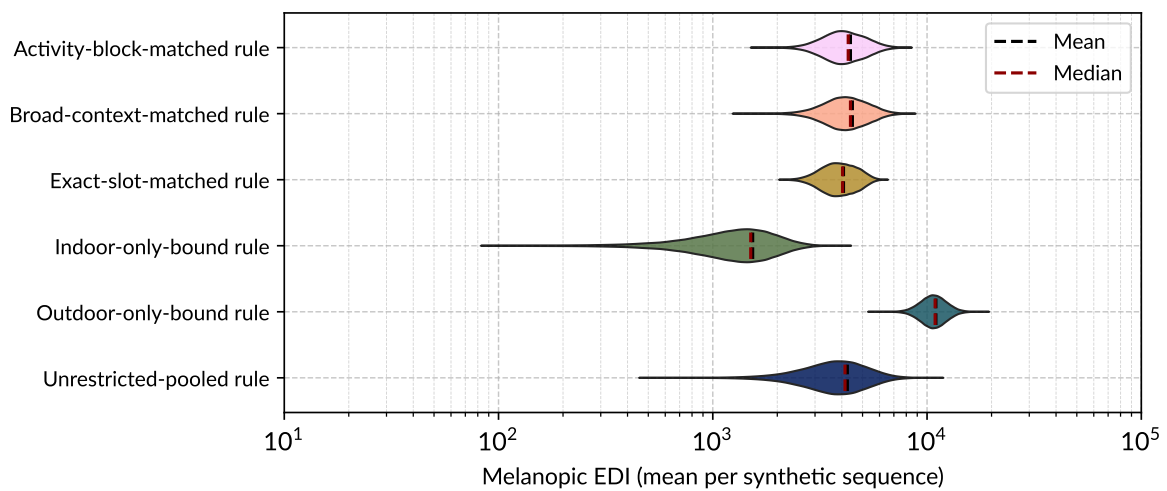
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Author comment: *Plot recreated locally with Python/R.*



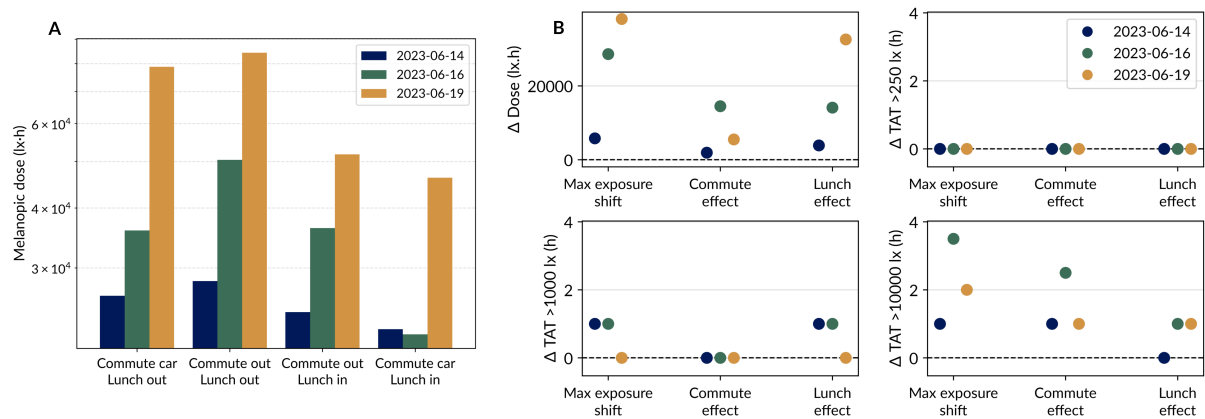
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Author comment: *Plot recreated locally with Python/R.*



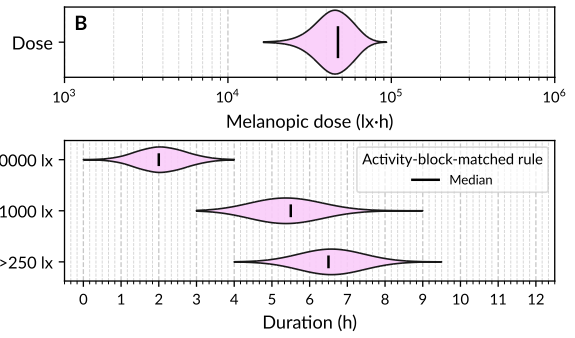
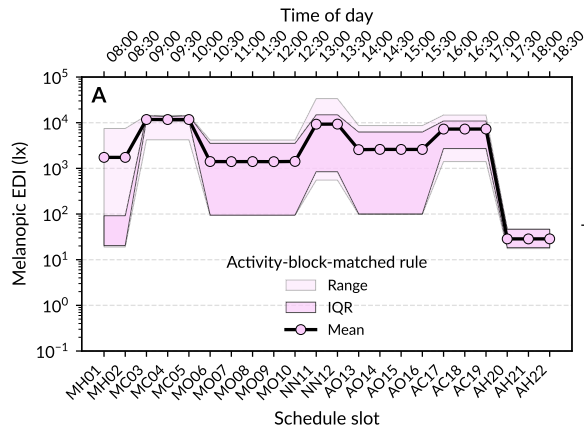
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Author comment: *Plot recreated locally with Python/R.*



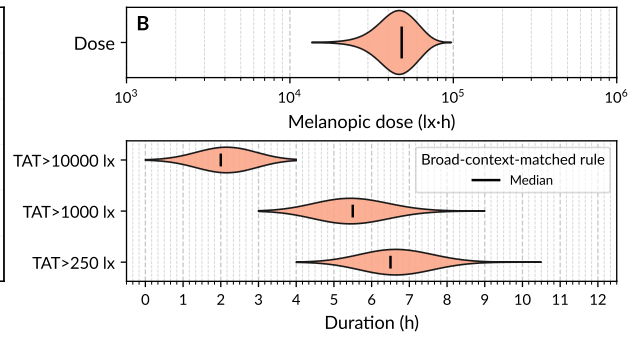
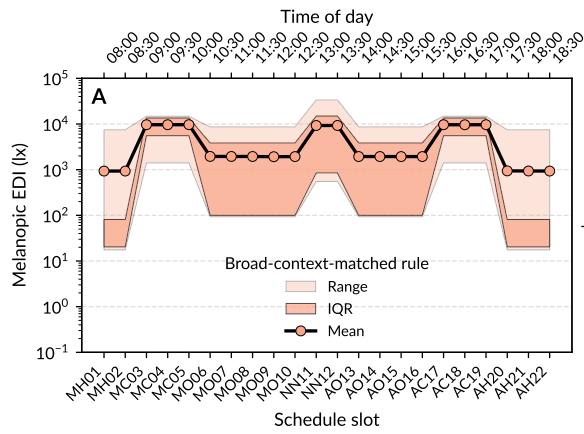
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Author comment: *Plot recreated locally with Python/R.*



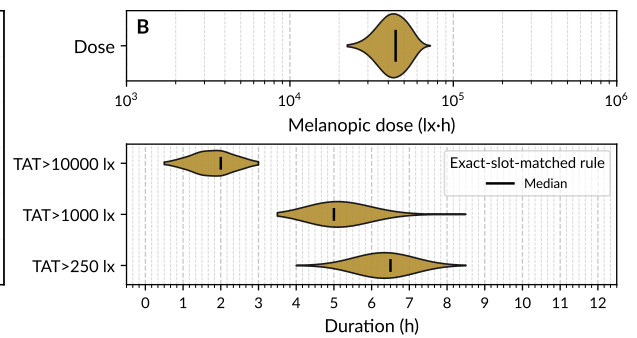
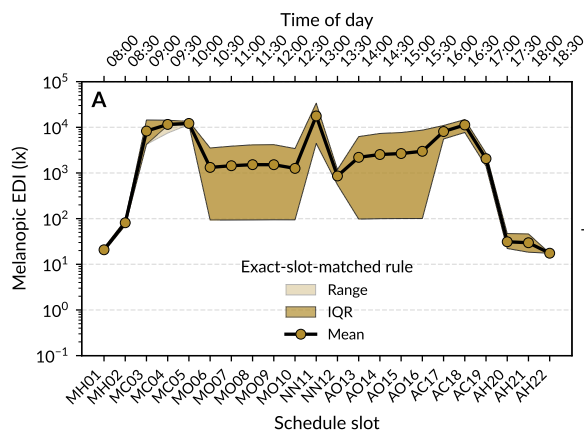
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Author comment: Plot recreated locally with Python/R.



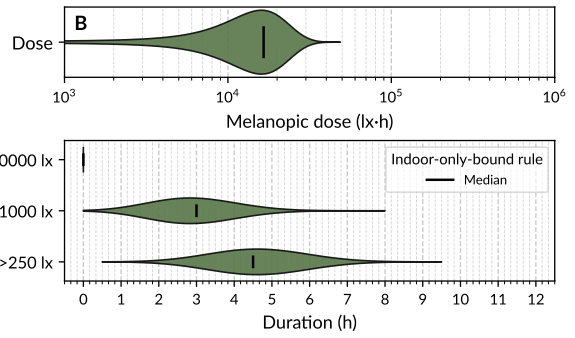
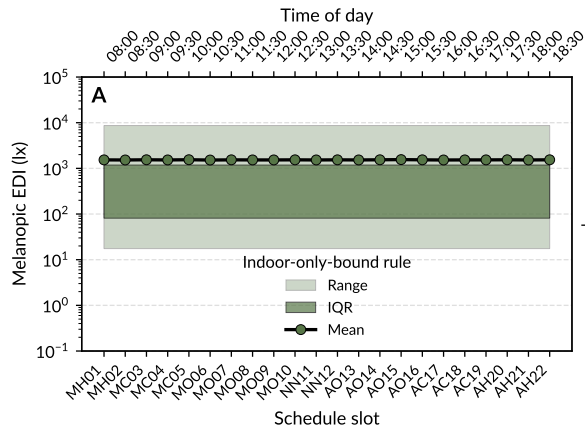
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Author comment: Plot recreated locally with Python/R.



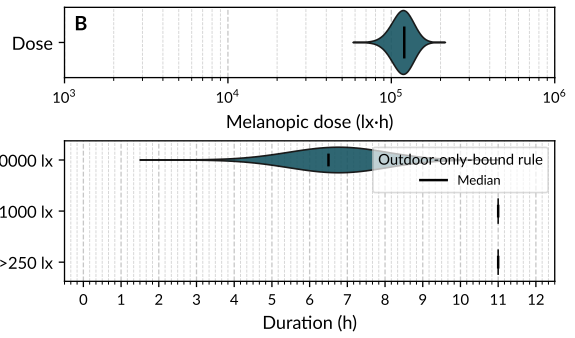
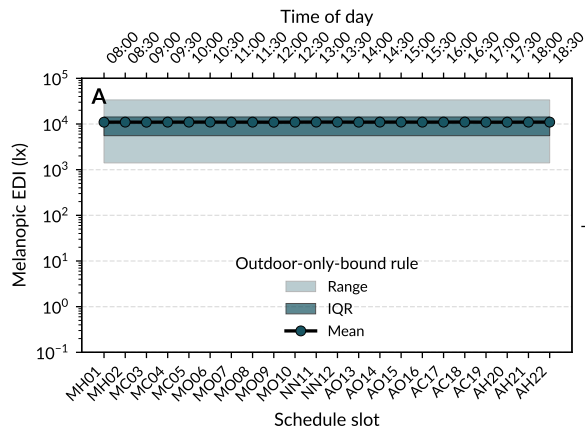
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Author comment: Plot recreated locally with Python/R.



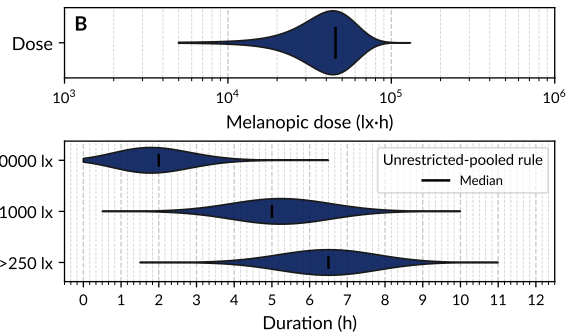
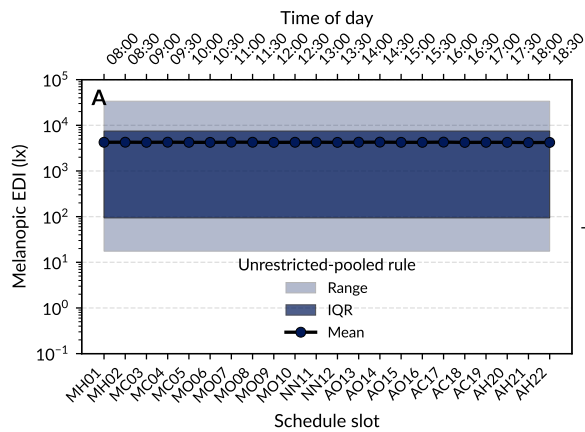
results/plots/outdoor_only_bound.pdf

Author comment: Plot recreated locally with Python/R.



results/plots/unrestricted_pooled.pdf

Author comment: Plot recreated locally with Python/R.



Acknowledgements

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Citing this document

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```
import session_info2 as si
si.session_info(os=True, cpu=True, gpu=True, dependencies=True)
```

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tornado 6.5.10
PyYAML 6.0.3
Brotli 1.2.0
wcwidth 0.9.1
colorama 0.4.6
traitlets 5.16.1
python-dateutil 2.9.0.post0
ipykernel 7.3.0
Pygments 2.21.0
pandas 3.0.6
packaging 26.3
idna 3.20
numpy 2.5.3
comm 0.2.3
certifi 2026.7.22 (2026.07.22)
charset-normalizer 3.5.1
platformdirs 4.11.12
ipython 9.17.1
PySocks 1.7.1
tabulate 0.10.0
pyzmq 27.2.0
jupyter_core 5.9.1
psutil 7.2.2
requests 2.34.2
urllib3 2.8.0
session-info2 0.4.2
debugpy 1.8.22
typing_extensions 4.16.0
jupyter_client 8.10.0
six 1.17.0
----
```

```
Python 3.14.7 | packaged by conda-forge | (main, Sep  2 2026, 21:08:32) [GCC 15.3.0]
OS Linux-7.0.0-34-generic-x86_64-with-glibc2.43
CPU 16/16 logical CPU cores
GPU No GPU found
Updated 2026-10-01 21:19
```

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